Unit 6 File Permissions

Course code LX13
Linux Basics

Objectives

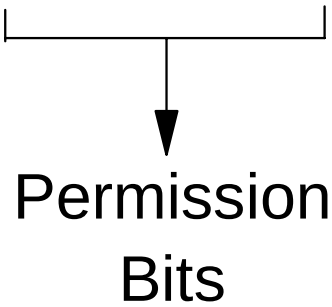
After completing this unit, students should be able to:

- List the basic file permissions
- Change the basic file permissions using both symbolic and octal formats

Long Listing of Files

- The **ls** command with the **-l** option can be used to obtain more information about the files in a directory.

```
$ ls -l
drwxrwxr-x  2 tux1 staff  1024  Aug 12  10:16  c
-rwxr-xr-x  2 tux1 staff  1024  Feb 18  09:55  doc
-rw-rw-r--  1 tux1 staff   767  Mar 24  23:34  mbox
```



File Protection/Permissions

For an ordinary file:

`r`	`read`	can look at the contents of the file
`w`	`write`	can change or delete the contents of a file
`x`	`execute`	can execute the file as a command (r also needed)

For a directory:

`r`	`read`	can find out what files are in directory
`w`	`write`	can create and remove files from directory (x also needed)
`x`	`execute`	can be the active directory

```
drwxrwxr-x  2 team01 staff  1024  Aug 12  10:16  c
-rwxr-xr-x  2 team01 staff  1024  Feb 18  09:55  doc
```

Changing Permissions Using Symbolic Notation

- The `change mode` command:

```
$ chmod mode filename
```

u = owner of file
g = group
o = other users on the system
+ = add permission
- = remove permission
= = clear permissions and set to mode specified

Symbolic Notation Example

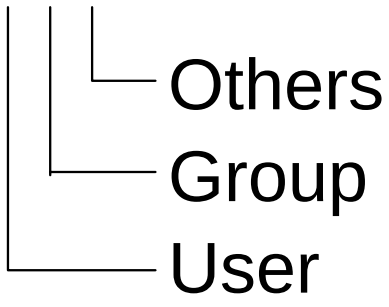
```
$ ls -l doc
-rwxr-xr-x  2 tux1 staff  1024  Feb 18  09:55  doc

$ chmod go-x doc
$ ls -l doc
-rwxr--r--  2 tux1 staff  1024  Feb 18  09:55  doc

$ chmod u-x,go+w doc
$ ls -l
-rw-rw-rw-  2 tux1 staff  1024  Feb 18  09:55  doc
```

Changing Permissions Using Octal Notation

- File and directory permissions can be specified in the symbolic syntax or as an octal number:

	User	Group	Other
Symbolic notation	rwx	rw-	r-x
Binary	111	110	101
	4+2+1	4+2+0	4+0+1
Octal	7	6	5

```
$ chmod 765 file
```



Octal Notation Example

```
$ ls -l doc
-rwxr-xr-x  2 tux1 staff  1024  Feb 18  09:55  doc

$ chmod 744 doc
$ ls -l doc
-rwxr--r--  2 tux1 staff  1024  Feb 18  09:55  doc

$ chmod 666 doc
$ ls -l
-rw-rw-rw-  2 tux1 staff  1024  Feb 18  09:55  doc
```

Default File Permissions

- The default permissions for newly created files and directories are:

File	`-rw-rw-r--`	`664`
Directory	`drwxrwxr-x`	`775`

- These default settings may be changed by changing the umask value, but it is usually sufficient to modify each new file or directory with special protection requirements using the `chmod` command

umask

- **umask** specifies what permission bits will be set on a new file or directory when created. It is an octal number that is used to determine what permission bits a file or directory is created with.

- The default value of 002 is set in **/etc/profile**. It can be changed for all users or for a specific user.

Write Permission on a Directory

```
$ ls -ld /home/tux1
drwxrwxrwx 2 tux1 tux1 1024 Apr  6  9:40 team01

$ ls -l /home/tux1/secret
-r--r--r-- 1 tux1 tux1 500 Apr 7 10:39   secret
```

As tux2 in /home/tux2:

```
$ echo "I removed your secret. Hahaa." > my.secret

$ mv my.secret /home/team01/secret
mv: replace secret, overriding mode 0444?

$ cat /home/team01/secret
I removed your secret. Hahaa.
```

Functions/Permissions Required

Command	Source Dir	Source File	Target Dir
cd	x	N/A	N/A
ls	r	N/A	N/A
ls -l	r,x	N/A	N/A
mkdir	w (parent) x	N/A	N/A
rmdir	w (parent) x	N/A	N/A
cat,pg,more,less	x	r	N/A
mv	w,x	NONE	w,x
cp	x	r	w,x
touch	w[1],x	N/A	NONE
rm	w,x	NONE	N/A

Checkpoint

The following questions are for a file called reportA which has the following permissions: `rw-r-----`

1. What is the mode in octal?

_____

2. Change the mode to `rwxrw-r--` using the symbolic format.

_____

3. Repeat step 2 but now using the octal format.

_____

Checkpoint *(2)*

4. Question 4 is based on the following listing.

```
$ ls -lR
Total 8
drwxr-xr-x 2 judy finance 1024 June 5  11:08 jobs

./jobs:
-rw-rw-r-- 1 judy finance  100 June 12 12:16 joblog
```

Can fred, who is member of the finance group, modify the file **joblog**?

Checkpoint *(3)*

5. Question 5 is based on the following listing.

```
$ ls -lR
Total 8
drwxrwxr-x 3 judy finance 1024 june 5 11:08 jobs

./jobs:
Total 8
drwxrw-r-x 2 judy finance 1024 june 7 11:10 work

./jobs/work:
Total 8
-rw-rw-r-- 1 judy finance 100 june 9 16:59 joblog
```

Can fred, who is a member of the finance group, modify the file joblog?

Checkpoint *(4)*

6. Question 6 is based on the following listing.

```
$ ls -lR
Total 8
drwxr-xr-x 3 judy finance 1024 june 2 11:08 jobs

./jobs:
Total 8
drwxrwxrwx 2 judy finance 1024 june 5 11:10 work

./jobs/work:
Total 8
-rw-rw-r-- 1 judy finance 100 June 6 12:16 joblog
```

Can Fred, who is member of the finance group, copy the file joblog to his home directory?

Unit Summary

- Basic file permissions can be listed using the ls -l command
- `chmod` grants or removes read, write and execute permissions for three classes of users: user, group and others
- The permissions used with the chmod command can be defined in symbolic or octal format